

Concept 1 | Fundamentals of Random Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Expectation

$$E(X) = \sum x p(x)$$

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

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Q001 **Written**

For the sum of two dice, construct the probability distribution and find $P(5 \leq X \leq 8)$.

Q002 Written

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

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Q003 **Written**

Construct both given probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Q004 Written

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

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Q005 Written

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Q006 Written**Find EX for all six given random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).**

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Q007 Written

For tails in four fair coins, find $E(3X+2)$, $E(-4X+1)$, and EX^2 .

Q008 Written

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

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Q009 **Written****Solve both given Exercise groups of transformed expectations for coins and for balls.**

Q010 Written

For black balls drawn from the stated box, find EX and VX .

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Q011 Written

Find the mean and variance for the two given Try random variables (a numbered card and winning tickets).

Q012 Written**Find EX and VX for all five given Exercise random variables.**

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Q013 Written

Find the standard deviation of tails when two biased coins have $P(\text{head})=4/7$ and $P(\text{tail})=3/7$.

Q014 Written

Find the standard deviation from each given probability table and for white balls drawn from 4 red and 3 white.

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Q015 **Written****Find the standard deviation for both given probability-table exercises.**

Q016 Written

For the number of winning tickets in the given draw, find VY and σY for $Y=2X+1$ and $Y=-3X-2$.

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Q017 **Written**

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X.

Q018 Written

For a single die, find VY and σY for $Y=X+2$, $Y=3X+1$ and $Y=-2X+4$.

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Q019 Written

Solve the two given transformation problems: a net prize after an entry fee, and finding a, b from $E(aX+b)=7/3$ and $\sigma(aX+b)=3$.

Q020 Written**Solve the two given Try problems involving net winnings and determining a,b for a die transformation.**

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Q021 Written

Solve all four **given** transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Q022 MCQ**A probability distribution must have probabilities that sum to:****Choose the correct option.****A** 0**B** 1**C** the number of outcomes**D** the mean

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Q023 MCQ

For two fair dice, $PX = 2$ is:**Choose the correct option.**A $1/6$ B $1/36$ C $2/36$

D 0

Q024 MCQ**The number of heads in three coins can be:****Choose the correct option.****A** 1,2,3**B** 0,1,2,3**C** 0,3 only**D** any integer

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Q025 MCQ

The expected value is calculated as:**Choose the correct option.**

- A $\sum PX = x$
- B $\sum xPX = x$
- C largest x
- D variance squared

Q026 MCQ**The mean number of tails in four fair coins is:****Choose the correct option.****A** 1**B** 2**C** 3**D** 4

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Q027 MCQ

For a fair die, EX is:**Choose the correct option.**

A 3

B 3.5

C 4

D 6

Q028 MCQ **$E(aX+b)$ equals:****Choose the correct option.**A aEX B $aEX+b$ C $EX+b$ onlyD $a+b$

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Q029 MCQ

In general, EX^2 is:**Choose the correct option.**A $E[X]^2$ B equal to EX

C a weighted square sum

D always zero

Q030 MCQ**If $EX=2$, then $E(3X+2)$ is:****Choose the correct option.****A** 6**B** 8**C** 10**D** 12

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Q031 MCQ

The variance identity is:**Choose the correct option.**

A $E[X] - E[X^2]$

B $E[X^2] - E[X]^2$

C $E[X]^2$

D $\sqrt{V}$

Q032 MCQ**Variance can never be:****Choose the correct option.**

- A positive
- B zero
- C negative
- D fractional

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Q033 MCQ

The first step in a variance problem is to:**Choose the correct option.**

- A draw the distribution
- B take a square root
- C ignore probabilities
- D add the values

Q034 MCQ

Standard deviation is:**Choose the correct option.**

- A VX
- B $\sqrt{V(X)}$
- C VX squared
- D EX

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Q035 MCQ

Standard deviation has units:**Choose the correct option.**

A squared units

B the same units as X

C no units

D units cubed

Q036 MCQ **$V(aX+b)$ equals:****Choose the correct option.****A** $aVX+b$ **B** a squared times VX **C** $|a|VX$ **D** $VX+b$

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Q037 MCQ

 $\sigma(aX+b)$ equals:**Choose the correct option.**A $a \sigma X$ B $|a| \sigma X$ C $a^2 \sigma X$ D $\sigma X+b$

Q038 MCQ**For $Y=X+3$, the variance is:****Choose the correct option.****A** $VX+3$ **B** VX **C** $3VX$ **D** $9VX$

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Q039 MCQ

A net prize after a fixed fee is modeled as:**Choose the correct option.**

A $Y = X + b$

B $Y = aX + b$

C $Y = X^2$

D $Y = a + b$ only

Q040 MCQ**If $a > 0$ and $\sigma(aX+b)=3$, then a is:****Choose the correct option.****A $3\sigma X$** **B $3/\sigma X$** **C $\sigma X/3$** **D $3+b$**

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Q041 MCQ

Changing b in $aX+b$ changes:**Choose the correct option.**

- A variance only
- B standard deviation only
- C the mean only
- D nothing

Q042 MCQ**If $EX=2$ and $EX^2=7$, then VX is:****Choose the correct option.****A** 3**B** 5**C** 7**D** 9

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Quiz WRITTEN 1 **Concept Quiz · Written**

Find EX and VX for the number of heads in three fair coins.

Quiz WRITTEN 2 Concept Quiz · Written

For $Y=4X-1$ with $VX=2$, find VY and σY .

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Quiz WRITTEN 3 Concept Quiz · Written

For a fair die, find $E(2X+5)$.

Quiz WRITTEN 4 Concept Quiz · Written

If $\sigma X=2$ and $EX=3$, find $\sigma(-5X+7)$.

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Quiz MCQ 5 [Concept Quiz](#) · MCQ

If $VX=5$, $V(3X+1)$ is:

Choose the correct option.

A 5

B 8

C 15

D 45

Quiz MCQ 6 Concept Quiz · MCQ**For a fair die EX is:****Choose the correct option.****A** 2.5**B** 3**C** 3.5**D** 4

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Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

A probability table total must be:

Choose the correct option.

A 0

B 1

C 2

D the mean

Quiz MCQ 8 Concept Quiz · MCQ

If $\sigma_X=4$, $\sigma(X-9)$ is:

Choose the correct option.

A 0

B 4

C 13

D 16

Answer key

Use the question number shown on each page.

- Q001** Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; $P5 \leq X \leq 8 = 20/36 = 5/9$.
- Q002** Use the sample space or combination to list every possible X and sum the relevant probabilities.
- Q003** Use binomial or hypergeometric counts, then add probabilities for the requested values.
- Q004** For four fair coins $EX=2$; for the hypergeometric draw $EX=4/3$.
- Q005** Use $EX = \sum xPX = x$ after constructing each distribution.
- Q006** Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.
- Q007** Use $E(aX + b) = aEX + b$; calculate EX^2 directly from the table.
- Q008** Apply linearity for $aX+b$ and use the weighted square sum for X^2 .
- Q009** For every expression use $E(aX + b) = aEX + b$; evaluate powers from the distribution table.
- Q010** $EX = 6/7$; $VX = 20/49$.
- Q011** Construct each probability table, then use $VX = EX^2 - EX^2$.
- Q012** Use the distribution table for each experiment and the variance identity.
- Q013** The table gives $VX = 24/49$, so $\sigma X = 2 \sqrt{6/7}$.
- Q014** Use $\sigma X = \sqrt{V(X)}$ after evaluating the variance from the table.
- Q015** Compute EX , EX^2 , then $\sigma X = \sqrt{E(X^2) - E(X)^2}$.
- Q016** Use $V(aX + b) = a^2 VX$ and $\sigma(aX+b) = |a| \sigma X$.
- Q017** Apply the squared multiplier to VX and the absolute multiplier to σX .
- Q018** Use the variance and standard-deviation transformation formulas for each Y .
- Q019** Use $E(aX + b) = aEX + b$ and $\sigma(aX+b) = |a| \sigma X$, with $a > 0$.
- Q020** Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.
- Q021** Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.
- Q022** B
- Q023** B
- Q024** B
- Q025** B
- Q026** B
- Q027** B
- Q028** B
- Q029** C
- Q030** C
- Q031** B
- Q032** C
- Q033** A
- Q034** B
- Q035** B
- Q036** B
- Q037** B
- Q038** B
- Q039** B
- Q040** B
- Q041** C
- Q042** A
- Quiz WRITTEN 1** $EX=3/2$ and $VX=3/4$
- Quiz WRITTEN 2** $VY=32$ and $\sigma Y=4 \sqrt{2}$
- Quiz WRITTEN 3** 12
- Quiz WRITTEN 4** 10
- Quiz MCQ 5** D
- Quiz MCQ 6** C
- Quiz MCQ 7** B
- Quiz MCQ 8** B